

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. CHB-504: Physical Chemistry-III

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Write the BET equation and identify each term involved in it.
- (b) What is meant by the ionic strength of a solution?
- (c) Explain why there is a depression in freezing point of a solvent upon addition of a non-volatile solute.
- (d) State the Grotthuss-Draper law and Stark-Einstein law of photochemistry.
- (e) What is quantum yield? Write the reasons for high and low quantum yields.

Question 2.

- (a) What is the photostationary state? Deduce the Stern-Volmer equation and show how the value of the quenching constant and other constants could be determined. [8]
- (b) Derive the rate law for a unimolecular surface reaction on a solid catalyst (Langmuir-Hinshelwood mechanism). [6]

Question 3.

- (a) Discuss the influence of high-frequency AC on the conductance of electrolyte solutions (Falkenhagen effect). [6]
- (b) Explain the reduction in ionic mobility due to 'relaxation' and 'electrophoretic' effects. Why do these effects become significant with increasing concentration? [8]

Question 4.

- (a) Derive the Debye-Hückel-Onsager equation for the conductance of a strong electrolyte solution at a given concentration. Write the limiting forms. [8]
- (b) Discuss the primary and secondary processes in photochemical reactions. [6]

Question 5. Write short notes on any two of the following: [2 × 7 = 14]

- (a) Photosensitized reactions
- (b) Adsorption isotherms (Langmuir vs. Freundlich)
- (c) Wien effect on conductance under high electric fields